UNIVERSIDAD NACIONAL MAYOR DE SAN MARCOS

Faculty of Systems Engineering and Informatics



DOCTORAL PROGRAM IN DEEP TECH, ARTIFICIAL INTELLIGENCE & EMERGING TECHNOLOGIES

Research Methods and Scientific Integrity in AI and Advanced Technologies

DELIVERABLE No. 7

MODEL CARD & DATASET DATASHEET

Human-in-the-Loop Multimodal AAC Prototype in Spanish

COURSE INSTRUCTOR	DOCTORAL STUDENT
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2026 - II | LIMA, PERU

Evidence-status note: this is a research-stage documentation package. No participant data have been collected and no thesis performance results are claimed.

Deliverable 07

Documentation Package

Model Card + Dataset Datasheet for the doctoral AAC research

Submission scope

This report documents the planned research artifact and planned dataset using the nine-section Model Card structure and seven-section Datasheet lifecycle structure required in Session 7. It explicitly distinguishes current evidence from future evaluation: no participant data have been collected and no thesis performance results are claimed.

1. Research Artifact and Evidence Status

Research title. A Human-in-the-Loop Multimodal AAC Prototype in Spanish to Support Independent Functional Communication in Autistic Children Requiring Very Substantial Support.

Current status. Pre-implementation / pre-human-data-collection research specification. The protocol plans a bounded Design Science Research build–evaluate–iterate study; the evaluation design and dataset remain to be finalized before ethics approval and data collection.

Integrity boundary. No Wisconsin Breast Cancer classroom result is used as evidence for this research. All unresolved empirical fields remain explicitly marked as unknown or not yet collected.

2. Model Card — HITL Multimodal AAC-ES v0.0

2.1 Model details

Field	Documented value
Name / version	HITL Multimodal AAC-ES v0.0 — research specification

Field	Documented value
Owner / contact	Miguel Ángel Patiño Antonioli: UNMSM Doctoral Program in Deep Tech, AI & Emerging Technologies
Documentation date	21 October 2026
Development status	Research-stage prototype specification; not yet trained, clinically validated, or deployed.
Architecture	Planned multimodal AAC artifact integrating photographs/pictograms, text, synthesized speech, contextual information, interaction history, adaptive suggestion/ranking, and human-in-the-loop review. The child retains final control to select, reject, modify, ignore, or stop.
License	UNKNOWN: to investigate before any public release.

2.2 Intended use

Field	Documented value
Primary intended uses	Research prototype to support independent functional communication in Spanish for a bounded subgroup of minimally speaking or nonspeaking autistic children requiring very substantial support, within one structured therapeutic communication context.
Intended users	Research participants supported by

Field	Documented value
	caregivers and qualified professionals; researchers evaluating the artifact under an approved protocol.
Out-of-scope / misuse	NOT for autism diagnosis, clinical or medical recommendations, inference of thoughts or internal states, autonomous communication on behalf of the child, unsupervised deployment, or replacement of AAC/clinical professionals.

2.3 Factors

Field	Documented value
Relevant groups / conditions	Communication profile; functional communication repertoire; sensory/contextual needs; Spanish-language and Peruvian sociocultural context; activity/context; caregiver/specialist configuration; individual interaction history.
Evaluation factors	Planned disaggregation by participant and experimental condition (intelligent prototype vs predefined static AAC comparator). Where feasible, results will also be examined by communicative function and prompting level.

2.4 Metrics

Field	Documented value
Primary outcome 1	Independent functional communication acts per session.
Primary outcome 2	Communicative diversity: range of communicative functions or concepts expressed.
Primary outcome 3	Assistance / prompting level required to communicate.
Secondary / process measures	Message-selection time; rejected or modified AI suggestions; treatment fidelity; inter-observer agreement; focused caregiver/specialist feedback.
Variation approach	Repeated within-participant comparisons across sessions. Report participant-level trajectories and condition-wise summaries. For stochastic ML benchmarking, use mean ± std over at least 5 seeds, confidence intervals, and paired comparisons when appropriate.

2.5 Evaluation data

Field	Documented value
Dataset / design	NOT YET COLLECTED. Planned pilot evaluation: minimum four participants from one clearly defined subgroup; one structured

Field	Documented value
	therapeutic communication context; alternating-treatments single-case design comparing the intelligent prototype with a predefined static AAC condition.
Evaluation protocol	Conditions counterbalanced or randomized where feasible; repeated measurement across sessions; prespecified session procedures, prompting hierarchy, fidelity checklist, safety/stopping rules, coding rules, and analysis logic.
Preprocessing	Planned pseudonymization and session-level coding; identifiers stored separately; a prespecified subset independently double-coded. Exact preprocessing pipeline is UNKNOWN: to investigate and freeze before data collection.

2.6 Training data

Field	Documented value
Datasets	No thesis participant data have been collected or used for model training as of this deliverable. The final foundation model/provider and any fine-tuning or adaptation data are UNKNOWN: to investigate and document before implementation.

Field	Documented value
Personalization data	If used, personalization will rely only on authorized, minimized interaction history and configured preferences; it must not silently convert test/evaluation interactions into training data.
Known limitations	No final training corpus, model checkpoint, data license, or multilingual/cultural coverage assessment is yet available.

2.7 Quantitative analyses

Field	Documented value
Overall results	NO THESIS RESULTS YET, no participant-level or model-performance claims are made in Deliverable 7.
Disaggregated results	Planned disaggregated reporting: per-participant and per-condition trajectories, with prompting/assistance explicitly separated. Aggregate summaries will not replace individual trajectories.
Statistical interpretation	Quantitative results will be treated as bounded pilot evidence rather than population-level clinical effectiveness. Repeated within-participant comparisons will be emphasized.

2.8 Ethical considerations

Field	Documented value
Risks	Privacy and security risks involving minors; unsafe or contextually inappropriate suggestions; bias; homogenization of communication; over-personalization; dependency; misinterpretation; reduction of user agency; inadvertent disclosure of sensitive routines or communication content.
Mitigations	Caregiver informed consent; communication-appropriate assent/withdrawal signals; data minimization; restricted access; pseudonymization; preference for on-device processing where feasible; human review; traceability of suggestions/modifications; ability to reject/ignore/stop; low-confidence blocking; bias/error review; no diagnosis or medical advice.
Human agency principle	AI outputs are suggestions, never decisions. The system must not speak autonomously on behalf of the child.

2.9 Caveats and recommendations

Field	Documented value
Caveats	Research-stage specification only. No clinical effectiveness, safety, generalizability, subgroup performance, external validation, or production reliability

Field	Documented value
	has been established.
Recommendations	Freeze the artifact version, comparator, eligibility criteria, model/provider, data schema, preprocessing, prompting hierarchy, coding rules, stopping rules, statistical plan, retention/deletion policy, and software environment before human-data collection. Version every DSR change and justify deviations.

3. Dataset Datasheet — Planned AAC Interaction Dataset v0.0

3.1 Motivation

Field	Documented value
Purpose	Create a traceable research dataset for evaluating whether a human-in-the-loop, context-aware multimodal AAC prototype supports independent functional communication more effectively than a predefined static AAC condition.
Created by / funded by	Planned doctoral research dataset led by Miguel Ángel Patiño Antonioli at UNMSM. Data-collection funding source: UNKNOWN: to investigate/confirm before collection.

3.2 Composition

Field	Documented value
What instances represent	Planned repeated communication opportunities / session-level interaction records linked to a pseudonymous participant and experimental condition.
Core fields	Pseudonymous participant ID; condition; activity/context; system suggestions; selection/rejection/modification; coded functional communication act and communicative function/concept; assistance/prompting level; message-selection time; fidelity/coding metadata; optional structured caregiver/specialist feedback.
Planned sample vs population	Minimum four participants from one clearly defined subgroup of minimally speaking or nonspeaking autistic children requiring very substantial support. This is a bounded pilot and is NOT representative of all autistic children.
Current status / missingness	NO HUMAN DATA COLLECTED YET. Final sample size, session count, exact field dictionary, missing-data rules, and subgroup coverage are UNKNOWN: to investigate before ethics approval and collection.

3.3 Collection process

Field	Documented value
How collected	Planned collection in one structured therapeutic communication context through prototype interaction logs, direct coding of functional communication acts, prompting/assistance records, rejected or modified AI suggestions, treatment-fidelity checks, and focused caregiver/specialist feedback.
Experimental structure	Leading design: alternating-treatments single-case comparison of the intelligent prototype versus a predefined static AAC comparator, with repeated measurement and counterbalanced/randomized condition order where feasible.
Consent / ethics	Collection will begin only after ethics approval. Written caregiver consent and developmentally/communication-appropriate assent procedures are required; observable withdrawal or distress signals will stop participation.
Timeframe / sites	Future collection; exact dates, recruitment site(s), and access arrangements are UNKNOWN: to investigate and document before recruitment.

3.4 Preprocessing / cleaning / labeling

Field	Documented value
Steps	Planned: separate direct identifiers from research records; assign pseudonymous IDs; validate session timestamps/condition labels; apply prespecified coding rules for independent vs prompted acts, communicative diversity, and assistance level; independently double-code a prespecified subset; document protocol deviations and fidelity.
Leakage control	Evaluation sessions must remain analytically separated from any model adaptation or personalization used for training. Any personalization update rule will be frozen and logged so future/test information cannot silently enter model development.
Raw data available	Identifiable photographs, voice-related data, and raw interaction records involving minors will NOT be placed in a public repository. Public release, if any, will be limited to ethically permitted de-identified/aggregate artifacts.

3.5 Uses

Field	Documented value
Appropriate uses	Doctoral evaluation of the bounded AAC prototype; reproducibility/audit of the

Field	Documented value
	planned comparison; analysis of functional communication outcomes, user agency, system failure modes, and artifact refinement.
Should NOT be used for	Autism diagnosis; medical/clinical prediction; autonomous inference of thoughts, emotions, pain, or intent; profiling beyond the approved research questions; training unrelated models; public identification of participants; broad population-level effectiveness claims.

3.6 Distribution

Field	Documented value
How distributed	Raw participant data are planned to remain access-controlled. Code, schemas, synthetic examples, documentation, and ethically permitted de-identified/aggregate outputs may be shared through the project repository after review.
License / restrictions	Final data-sharing license, repository access level, and data-use agreement are UNKNOWN: to investigate and align with ethics approval, institutional policy, and consent language.

3.7 Maintenance

Field	Documented value
Maintainer	Doctoral researcher / research team under UNMSM governance, subject to final protocol approval.
Versioning	Dataset schema, coding manual, artifact version, comparator, and DSR iteration changes will be versioned and justified. Protocol deviations will be logged.
Retention / deletion	Retention period, secure storage location, backup strategy, access revocation, and deletion schedule are UNKNOWN — to investigate and finalize before data collection.

4. Documentation Scorecard — Self-Audit

Result: 8 / 8

The scorecard evaluates documentation completeness and transparency, not model accuracy or clinical effectiveness. The 8/8 result is achieved without fabricating empirical outcomes: planned variation and disaggregation are stated, misuse is explicit, the collection process is described, and unresolved items are flagged honestly.

Check	Status	Interpretation
Model Card has all 9 sections	**PASS**	Documented in the Model Card / Datasheet.
Datasheet has all 7 sections	**PASS**	Documented in the Model Card / Datasheet.

Intended misuse / out-of-scope stated	**PASS**	Documented in the Model Card / Datasheet.
Performance reported with variation (± / std / seeds)	**PASS**	Documented in the Model Card / Datasheet.
Metrics disaggregated (per-class / per-group)	**PASS**	Documented in the Model Card / Datasheet.
Dataset 'should NOT be used for' stated	**PASS**	Documented in the Model Card / Datasheet.
Collection process described	**PASS**	Documented in the Model Card / Datasheet.
Gaps flagged honestly ('unknown')	**PASS**	Documented in the Model Card / Datasheet.

5. Educational EU AI Act Annex IV Crosswalk

This crosswalk is an educational gap analysis only; it is not a conformity assessment. Its value is to expose which technical-documentation sentences cannot yet be written because the research artifact is still pre-implementation.

Annex IV element	Model Card mapping	Status	Evidence / gap
IV(1) General description: purpose, provider, version	Model details; Intended use	**COVERED**	Purpose, owner, research-stage version, intended use and misuse are stated.
IV(2) Development process: data, design choices, methods	Training data; Evaluation data; Metrics	**PARTIAL**	Design choices and planned evaluation are documented, but final

			model/provider, training data, and human evaluation data are not yet available.
IV(3) Monitoring, functioning & control; accuracy by group	Factors; Quantitative analyses	**PARTIAL**	Planned participant/condition disaggregation is specified; no empirical subgroup results exist yet.
IV(4) Appropriateness of performance metrics	Metrics; Quantitative analyses	**PARTIAL**	Primary/secondary outcomes are prespecified, but empirical estimates and uncertainty are not yet available.
IV(5) Risk management: foreseeable risks & mitigations	Ethical considerations	**COVERED**	Foreseeable risks, human oversight, data minimization, traceability and agency safeguards are stated at research-protocol level.
IV(8) Post-market monitoring / known limitations	Caveats & recommendations	**PARTIAL**	Known research-stage limitations and pre-collection actions are stated; post-deployment monitoring does not

			yet apply.

6. Reproducibility and Pre-Collection Actions

- Freeze the artifact version and exact static AAC comparator before evaluation.
- Finalize the foundation model/provider, any fine-tuning/adaptation strategy, software versions, and model/data licenses.
- Finalize eligibility criteria, recruitment site(s), exact sample/session plan, and ethics/consent/assent procedures.
- Freeze the dataset schema, coding manual, prompting hierarchy, treatment-fidelity checklist, inter-observer agreement procedure, stopping rules, and protocol-deviation log.
- Prespecify the analysis plan, including participant-level trajectories, condition-wise comparisons, uncertainty, and any multiple-comparison correction.
- Separate evaluation data from model adaptation/personalization to prevent leakage; explicitly log any allowed personalization update.
- Finalize secure storage, retention, deletion, access-control, and public-release rules; do not publish identifiable data from minors.
- Version every Design Science Research change and document why it was made.

7. Submission Bundle

File	Purpose
model_card.md	Nine-section Model Card for the planned AAC artifact.
datasheet.md	Seven-section Datasheet for the planned AAC research dataset.
documentation_scorecard.md	8/8 self-audit based on Session 7 Exercise 6.
annex_iv_crosswalk.md	Educational documentation gap analysis.

REPRODUCIBILITY_REPORT.md	Evidence-status and reproducibility boundary.
Notebook_HandsOn_Executed [PATIÑO, Miguel]	Customized course notebook. Wisconsin exercises are explicitly educational only.
Session7_deliverable [PATIÑO, Miguel].zip	One-click bundle for submission / repository.

References

Mitchell, M., Wu, S., Zaldivar, A., et al. (2019). Model cards for model reporting. Proceedings of FAT* 2019, 220–229. arXiv:1810.03993.

Gebru, T., Morgenstern, J., Vecchione, B., et al. (2021). Datasheets for datasets. Communications of the ACM, 64(12), 86–92. arXiv:1803.09010.

Regulation (EU) 2024/1689 (Artificial Intelligence Act), Article 11 and Annex IV.

Patiño Antonioli, M. Á. (2026). Research Protocol — Initial Draft, Deliverable No. 3. UNMSM doctoral course document.

Patiño Antonioli, M. Á. (2026). Proyecto de investigación: Sistema inteligente multimodal de comunicación aumentativa y alternativa en español para mejorar la comunicación funcional de niños con trastorno del espectro autista y necesidades de apoyo muy sustanciales. UNMSM.